

Maher Mamah

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Research Summary

PhD researcher in Combinatorics & Optimization at the University of Waterloo specializing in post-quantum cryptography, quantum information processing, and algorithmic design. Research focuses on developing quantum-resistant cryptographic systems from isogenies and codes, mathematical foundations of secure computing, and cryptographic protocol analysis.

Education

2025–present	PhD in Combinatorics & Optimization	University of Waterloo, Canada Expected Graduation: May 2029
2023–2025	Masters in Mathematics	Pennsylvania State University, USA
2020–2023	B.Sc. in Applied Mathematics	American University of Beirut, Lebanon

Research Experience

PhD Researcher – Department of Combinatorics & Optimization

University of Waterloo, Canada

2025–Present

Research under the guidance of Prof. *David Jao and Sam Jaques*

- Conduct research in post-quantum cryptography, focusing on isogeny-based and code-based cryptographic constructions.
- Currently developing a code-based cryptographic protocol based on the LESS signature scheme, with the goal of reducing signature sizes by approximately 25% (*joint work with Dr.s Jao and Jaques*).
- Develop and evaluate cryptographic algorithms using Python, SageMath, C/C++ for real world key-exchange protocols like CSIDH.

Graduate Researcher – Department of Mathematics & Computer Science

Pennsylvania State University, USA

2023–2025

Research under the guidance of Prof. *Kirsten Eisenträger and Prof. Sean Hallgren*

- Conducted research on theoretical foundational problems in isogeny-based cryptography and their hardness relationships through security reductions.
- Investigated an open problem in quantum computing application to isogeny-based cryptography to devising a quantum algorithm for generating secure elliptic curves for protocols instantiation.

Publications & Preprints

1. **Maher Mamah, David Jao.** *A Note on the Security Proof of SQIsign*. Accepted to ACM Conference on Post-Quantum and Quantum Security 2026.
2. **Leonardo Colo, Maher Mamah, Youcef Mokrani, Bruno Sterner, Nicolas Swanson.** *Zero-Knowledge Proofs of Isogeny Diamonds*. Accepted to Communications in Cryptology journal. [ePrint](#).
3. **Maher Mamah.** *Complexity Analysis and Security Implications of the new Isogeny-Path algorithm*. Submitted to Public Key Cryptography PKC 2027.
4. **Maher Mamah, Jake Doliskani, David Jao.** *On the Spectral Theory of Isogeny Graphs and Quantum Sampling of Secure Supersingular Elliptic Curves*. Submitted to EUROCRYPT [ePrint](#).
5. **Maher Mamah.** *Enhanced Algorithms for the Representation of Integers by Binary Quadratic Forms: Reduction to Subset Sum*. [arXiv](#).
6. **Maher Mamah.** *The Supersingular Isogeny Path and Endomorphism Ring Problems: Unconditional Reductions*. Communications in Cryptology Journal, vol. 3, no. 2, 2026. [Article](#).

7. **Maher Mamah, A. Saraeb.** *A Generalization of Siegel's Method to Jacobi's Function.* *The Ramanujan Journal*, 65, 521–531 (2024).
8. **M. Me'meh, A. Saraeb.** *A Generalization of Iseki's Formula and the Jacobi Theta Function.* *Hardy-Ramanujan Journal*, 45, 130–139.

Talks & Presentations

July 2026	<i>Lattices, Isogenies and Codes: Contrasting Post-Quantum Design Directions for Applied Cryptography</i>	Cedarcrypt Conference, AUB Mediterranean Cyprus-Greece
May 2026	<i>Code-based Cryptography: The Decoding Problem and McEliece Encryption Scheme</i>	Code-based Cryptography School – UWaterloo
Nov 2026	<i>The Endomorphism Ring Problem and Applications to SQIsign Digital Signature</i>	Cryptography Reading Seminar – UWaterloo
Oct 2025	<i>Act2Qrypt Seminar: Isogeny-based Cryptography and the Underlying Hard Problems</i>	UWaterloo & UCalgary
March 2025	<i>Introduction to Post-Quantum Cryptography: From Elliptic Curves to Isogenies</i>	Penn State
March 2025	<i>Number Theoretic Aspects of Supersingular Isogeny Path and Endomorphism Ring Problems</i>	University of Maryland – MASON V

Honors and Awards

2026	William Tutte Scholarship in Research	University of Waterloo
2025	Graduate Research Studentship Award	University of Waterloo
2025	MASON V Conference Travel Grant	University of Maryland
2023	Jack and Eleanor Pettit Scholarship Grant in Science	Penn State
2022-2023	Qatar Foundation Scholarship Award	American University of Beirut
2020-2023	USAID Full Scholarship Fund	American University of Beirut

Teaching Experience

Fall 2026	Teaching Assistant of Applied Cryptography	UWaterloo
Winter 2026	Teaching Assistant of Applied Cryptography	UWaterloo
Fall 2025	Teaching Assistant of Calculus 1	UWaterloo
Summer 2025	Instructor of Integral Vector Calculus	Penn State
Spring 2025	Instructor of Calculus of Several Variables	Penn State
Fall 2024	Instructor of Calculus and Vector Analysis	Penn State
Spring 2023	Teaching Assistant of Calculus & Analytic Geometry	Penn State

Technical Skills

- **Programming Languages:** Python, MATLAB
- **Computational Tools:** SageMath, Mathematica, LaTeX
- **Research Areas:** Post-Quantum Cryptography, Quantum Information Science, Cryptographic Protocols, Algorithm Design, Computational Complexity.
- **Languages:** English (Fluent), Arabic (Native)